Operations with Fractions

1 Background Knowledge Check

1. See the "Introduction to Fractions" SLS document

2 Multiplying Fractions:

Multiplying fractions just requires us to multiply the numerators together and multiply the denominators together:

$$\frac{3}{2} \times \frac{7}{6} = \frac{21}{12}$$

Both 21 and 12 share a common factor of 3, therefore we simplify our fraction by dividing both the numerator and the denominator by 3:

$$\frac{21}{12} = \frac{21 \div 3}{12 \div 3} = \frac{7}{4}$$

Since 7 and 4 don't share any common factors other than 1, our final answer is $\frac{7}{4}$.

Tip: Always try to reduce fractions before multiplying; this makes doing the actual question much easier, as you are able to multiply smaller numbers together instead. For example:

$$\begin{aligned} &\frac{\cancel{72}^{8}}{\cancel{18}^{2}} \times \frac{3}{5} \\ &= \frac{\cancel{8}^{4}}{\cancel{2}^{1}} \times \frac{3}{5} \\ &= \frac{4}{1} \times \frac{3}{5} \\ &= \frac{12}{5} \end{aligned}$$

Before multiplying, you can cancel numbers diagonally if they share a factor. This works because that factor would end up on both the top and bottom anyway, so removing it early makes things easier:

$$\begin{aligned} &\frac{\cancel{64}^{8}}{3} \times \frac{5}{\cancel{24}^{3}} \\ &= \frac{8}{3} \times \frac{5}{3} \\ &= \frac{40}{9} \end{aligned}$$

3 Dividing Fractions:

When dividing fractions, we first flip the numerator and denominator of the fraction we are diving by and turn the division into multiplication. For example:

$$\frac{2}{5} \div \frac{6}{7}$$

$$= \frac{2}{5} \times \frac{7}{6}$$

It is important that only the fractions that we are dividing by is flipped, as flipping the fraction being divided will give us the wrong answer. Next, we just multiply as we did before:

$$\frac{2}{5} \times \frac{7}{6} = \frac{14}{30}$$

Now we once again reduce:

$$\frac{\cancel{14}^{7}}{\cancel{30}^{15}} = \frac{7}{15}$$

and done!

4 Adding and Subtracting Fractions

Fractions, just like whole numbers, can be added and/or subtracted from any other number. However, when working with fractions, we must make sure anything we are adding or subtracting from a fraction must also be a fraction and vice versa. Of course, we may have situations like as follows:

$$4 + \frac{1}{2}$$

To solve such questions, we must first start by turning any whole numbers into fractions. Any whole number can be turned into a fraction by dividing that number by 1. For example, 2 can be turned into the fraction $\frac{2}{1}$. So we start by turning the 4 in the question above into a fraction:

$$\frac{4}{1} + \frac{1}{2}$$

Our next step is to always make sure that any fractions we add or subtract have the same denominators. To achieve this we want to find the least common multiple between both of our current denominators (lowest numbers that both of the denominators multiply to). For our denominators 2 and 1, both of them can multiply to 2. So we call 2 our least common denomiator (LCD):

$$2 \times 1 = 2 \;\&\; 1 \times 2 = 2$$

Now that we know our least common multiple is 2, we need to multiply each denomintor with a number that results in 2, we must also multiply the numerator by that same exact number. For our fraction $\frac{4}{1}$ we must multiply the denominator by 2 for it be equal 2, so we do the same with the numerator:

$$\frac{4}{1} \times \frac{2}{2} = \frac{8}{2}$$

This gives us an equivalent fraction ($\frac{8}{2}$) to the one we started with ($\frac{4}{1}$). Lets repeat the same process for our other fraction of $\frac{1}{2}$:

$$\frac{1}{2} \times \frac{1}{1} = \frac{1}{2}$$

Notice that our fraction didn't change, since our fraction already had a denominator of 2, we could've just left it as is. Now let's input our new fractions into our equation:

$$\frac{4}{1} + \frac{1}{2} = \frac{8}{2} + \frac{1}{2}$$

Now, since both of our fractions have a common denominator; we can just add our numerators:

$$\frac{8}{2} + \frac{1}{2} = \frac{8+1}{2} = \frac{9}{2}$$

Since $\frac{9}{2}$ can't be reduced any further, we leave it as our final answer. The same process also applies to subtraction:

$$\begin{aligned} &4 - \frac{1}{2} \\ &= \frac{4}{1} - \frac{1}{2} \\ &= \frac{8}{2} - \frac{1}{2} \\ &= \frac{8-1}{2} \\ &= \frac{7}{2} \end{aligned}$$

Steps for Adding & Subtracting Fractions

1. Turn any whole numbers into fractions by dividing by 1
2. Find a least common denominator
3. Turn all fractions into equivalent fractions so that their denominator matches the least common denominator
4. Combine all the numerators into one while keeping the operations between them
5. Add or subtract the numbers in the numerator accordingly
6. Reduce the final answer if possible

Practice Problems

1. **Evaluate the following (Remember to always simplify your answer):**
   (a) $\frac{2}{3} + \frac{1}{3}$
   (b) $\frac{6}{5} - \frac{4}{4}$
   (c) $\frac{1}{7} + \frac{9}{2}$
   (d) $\frac{3}{2} - \frac{6}{4}$
   (e) $\frac{21}{3} - 2$
   (f) $1 + \frac{7}{14}$

2. **Evaluate the following (Remember to always simplify your answer):**
   (a) $\frac{10}{32} \times \frac{2}{1}$
   (b) $\frac{2}{9} \div \frac{1}{4}$
   (c) $\frac{8}{2} \div \frac{2}{14}$
   (d) $\frac{5}{10} \times \frac{10}{5}$
   (e) $\frac{22}{5} \times \frac{25}{9}$

3. **Simplify the following fractions as much as possible:**
   (a) $\frac{100}{25}$
   (b) $\frac{27}{15}$
   (c) $\frac{24}{23}$
   (d) $\frac{8}{64}$
   (e) $\frac{264}{1056}$
   (f) $\frac{81}{36}$
   (g) $\frac{21}{56}$

Solutions

1. Evaluate
(a) 1
(b) $\frac{1}{5}$
(c) $\frac{65}{14}$
(d) 0
(e) 5
(f) $\frac{3}{2}$

2. Evaluate
(a) $\frac{5}{8}$
(b) $\frac{8}{9}$
(c) 28
(d) 1
(e) $\frac{110}{9}$

3. Simplify
(a) 4
(b) $\frac{9}{5}$
(c) $\frac{24}{23}$
(d) $\frac{1}{8}$
(e) $\frac{1}{4}$
(f) $\frac{9}{4}$
(g) $\frac{3}{8}$